

Aditya Wadhwa

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EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Applied Science, Management Engineering - GPA (4.0)

Sept. 2025 – Present

- **Relevant Coursework:** Software Engineering, Data Structures & Algorithms, Systems Design, Optimization, Databases, Distributed Systems, Operating Systems, Machine Learning

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, HTML/CSS, SQL, R, Ruby, VBA

Frameworks & AI: React (DOM), Vite, Flask, Ruby on Rails, Llama API, LangChain, RAG, MCP, RandomForest

Libraries: pandas, NumPy, Scikit-learn, Matplotlib, OpenLayers, Three.js, millify, tkinter, yahooquery

Developer Tools: git/GitHub, Docker, Kubernetes, Power BI/Service, Tableau Desktop/Server, Claude Code

EXPERIENCE

Privy Council Office

Ottawa, ON

Data Analyst/Engineer

May 2026 – Aug. 2026

- Built interactive employment equity dashboards in Tableau Server and Power BI, regularly utilized by CSB directors and PCO executives to identify workforce representation gaps and monitor progress on inclusion goals.
- Reduced manual preparation time for finance (overtime, 10-15 year projection, monthly/quarterly) tracking reports by ~40% by automating recurring data cleaning and report generation workflows in Python and Power Query.
- Employed ETL operations with Power Query, extracting and transforming ministerial regional office data into elaborative dashboards and structured walkthroughs delivered to CSB executive directors and director generals.

PROJECTS

ExoDiscover | Python, RandomForest, Flask, Three.js, React

Nov. 2025

- Engineered a machine learning pipeline addressing NASA's on-going exo-planet detection challenge by training RandomForest classifiers on 50GB of Kepler/TESS/K2 mission datasets.
- Placed top 3 among 150+ participants by pitching a full-stack web application built on a Flask backend with interactive Three.js frontend visualizations.

AccessView | JavaScript, OpenLayers, OpenRouteService, Flask, Git

Sep. 2026

- Developed an interactive mapping application that helps mobility-limited and senior residents find accessible walking routes and nearby amenities (pharmacies, clinics, transit stops) by ingesting OpenStreetMap and municipal open-data GeoJSON into styled, category-toggable vector layers.
- Cut accessible trip planning to a two-click interaction by integrating the OpenRouteService API with wheelchair and foot-walking profiles, rendering live turn-by-turn GeoJSON routes and caching repeat requests to stay within free-tier limits at zero API cost.

NHL Game Predictor Pipeline | Python, Scikit-learn, Pandas, Git

Sep. 2026

- Built an end-to-end NHL game-outcome pipeline ensembling Logistic Regression and Gradient Boosting classifiers in Scikit-learn, reaching 56% test accuracy on a time-aware split against a ~55% home-ice baseline, with win probabilities calibrated via log-loss and Brier score.
- Automated ingestion and feature extraction in Pandas across 10,000+ historical match records, deriving prior-season team-strength features that prevent target leakage between seasons.

Mock Stock Portfolio | Python, YahooQuery API, NumPy, tkinter

Feb. 2025

- Created a real-time trading simulator enabling risk-free strategy testing by integrating the YahooQuery API with a Python and NumPy backend for high-speed data processing.
- Reduced asset research time by 45% by programming a dynamic stock screener that filters assets on P/E ratios, ESG scores, debt/return to equity ratios and growth metrics.

Financial Planning Excel Tool | VBA, Macros, Statistical Analysis

Dec. 2025

- Designed an automated budgeting tool improving financial literacy for first-year engineering students by scripting advanced VBA macros and interactive UserForms.
- Streamlined variance analysis by engineering comparison algorithms that visualize Actual vs. Ideal budget differences in real time.